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Digital Twins for Construction Sites: Concepts, LoD Definition, and Applications

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Abstract: The lack of integration between the digital and physical world results in a lower level of efficiency and collaboration in the construction industry. Digital twin technology, which creates a visual and digital model of a corresponding physical object for simulating, monitoring, analyzing, and other actions throughout the whole life cycle, is considered as an effective solution to address these problems. This research proposes a framework to utilize digital twins and extend the existing level of details (LoDs) of building information modeling (BIM) for construction site management. This study analyzes and improves the operation principle and mechanism of digital twins, including the digital representation based on BIM, Internet of Things (IoT), data storage, integration, and analytics, as well as interaction with the physical environment. Questionnaires and interview results verify that the proposed framework can support construction site monitoring and management, enhance quality and efficiency, and improve construction safety. It also acknowledges the contribution of LoDs' extension to construction site management. The interviews underline the main challenges that BIM, IoT, and data processes face in practical applications.
DOI: 10.1061/(ASCE)ME.1943-5479.0000948. *© 2021 American Society of Civil Engineers.*

Author keywords: Digital twins; Building information modeling (BIM); Construction sites; Level of details (LoDs).

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Introduction

In the construction industry, the lack of integration between the digital world and physical world, especially the interaction between data analysis and the corresponding action, has led to information fragments as well as data duplication and stagnation between different stages in the construction life cycle. It lowers the efficiency of management and hinders collaboration between staff members, diverse professions, and distinct construction phases. In particular, the situations on a construction site are dynamic with constant changes in facilities, equipment, materials, people, site layout, and various other components at different phases, which increases the requirement for coordination and collaboration (Kumar and Cheng 2015). Consequently, these increase the complexity of monitoring and management of construction sites.

Construction informatization has become the future direction of development, whereby data-driven and dynamic modeling methods are employed to address some concerns in the field of engineering management in recent years, such as worker-power management (Kim et al. 2020), safety monitoring (Su et al. 2021), and decision-making support (Abotaleb and El-adaway 2018) as well as workflow administration (Ma et al. 2018). The digital twin has the potential capability of providing the effective solutions to address these problems (Lu et al. 2020c).

Digital twin technology is defined as creating a visual and digital model of a physical object (Kaur et al. 2020). There is a profound interaction between the physical and digital components through the continual and dynamic data exchange between the physical and digital environments. It provides real-time monitoring, updating, simulating, analyzing, controlling, predicting, and optimizing (Kaur et al. 2020; Schleich et al. 2017), as well as validating and seamlessly coordinating throughout the life cycle. As a result, information silos and data duplication can be effectively avoided. By using this approach, the construction site situation can be monitored and managed alongside the whole process. The digital twin basic architecture contains (1) the digital representation based on building information modeling (BIM) technology, (2) Internet of Things (IoT) containing sensor technology, (3) data storage, integration and analytics, and (4) interaction with the physical environment.

Digital representation is the foundation of the digital twin. BIM is the key technology to build the digital representation because BIM establishes corresponding information models for the target components and collects information corresponding to components. Three-dimensional modeling of BIM for specific construction projects can provide stakeholders with a visualization function to show the overall situation of the project and its components (Davis 2003). Considering the limitations of hardware and software and the purpose of BIM usage in the different stages, it is not necessary to model BIM objects with 100% reality (CIC 2019d). As a result, the level of details (LoDs) can be utilized to determine the minimum detail level of BIM objects. The American Institute of Architects (AIA) defines LoDs as the measure, estimate, and schedule progression of a model (BIMForum 2016). The CityGML standard clarifies that an object has different representations for each LoD, which allows for analyzing and visualizing the same object with several degrees of resolution (Gröger and Plümer 2012).

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Note. This manuscript was submitted on October 23, 2020; approved on April 29, 2021; published online on December 14, 2021. Discussion period open until May 14, 2022; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Management in Engineering***, © ASCE, ISSN 0742-597X.

The extension of the LoDs definition supports and completes the BIM utilization, which facilitates the description of BIM used in BIM plans (Tolmer et al. 2017). In addition, sensor technology and sensor devices are employed to collect various real-time data in the whole life cycle (Lu et al. 2020c). Data storage, integration, and analytics are operated to capture, integrate, and analyze these sensor data. Such processes are conducted to obtain the required knowledge and realize added-value, supported by algorithmic analytics (Babiceanu and Seker 2016).

Some of the existing literature has indicated that construction-related applications of digital twins, and other studies are related to multiple standards and research on LoDs in the construction field. However, the literature review revealed that there is currently no existing research available that jointly considers the digital twin and LoDs in the construction sites management domain. The current research gaps are as follows: (1) there is no existing research that considers digital twins for the whole construction site process management and its comprehensive application for construction sites; (2) no standards or literature are available that define the LoDs of BIM for construction sites management and its intuitive model examples; and (3) no studies have yet explored the challenges faced by the key technologies of digital twins in practical applications and the critical views of managers in a transinternational background. As a result, this study proposes a framework to utilize digital twins for construction site monitoring and management and extends the existing LoDs of BIM for construction site management based on industry requirement analysis, a comprehensive literature review, and practical experience in related fields to bridge the research gaps.

The "Literature Review" section covers the previous studies, and analyzes and summarizes the application cases to determine the research gaps. The "Proposed Digital Twin Framework" section analyzes and outlines the operation principle and mechanism of the digital twin for construction sites, including digital representation; IoT; data storage, integration, and analytics; and interaction with the physical environment. For construction site monitoring and management, the digital twin technology contributes to realizing the integration of various construction site information, effectively managing and centralizing these information resources. As a result, the various requirements of construction sites at different stages can be monitored in real-time, which further provides decision-making support based on digital information and predicts the future development. The study expands the existing LoDs to construction sites for achieving BIM use expectation in the "LoDs of Digital Twins for Construction Sites" section. Ultimately, the framework proposed by this research can result in better construction site processes monitoring and management, including more efficiently responding to incidents, work efficiency enhancement, emission and energy performance optimization, safety, and quality promotion as well as equipment, assets, worker power, and supply chain administration improvement. Interviews and questionnaires are undertaken to gather the opinions from experts for validating the framework and supplementing the problems this framework might encounter in practical application in the "Validation" section. The "Future Directions" section points out some technical and nontechnical challenges for future study.

Literature Review

Digital twin technology provides a rich virtual representation of the physical environment (Kaur et al. 2020) for the entire life cycle. It can support the planning, tracking, monitoring, operating, risks and problems identifying, improvement, performance optimization, maintenance, and future resources requirement predicting (Kaur et al. 2020; Schleich et al. 2017). At present, this technology has been applied in many fields, including construction, industry, aviation, manufacturing, and health care as well as agriculture and others. Digital twins have been developed well in fields such as manufacturing. It guarantees support to manufacturing activities such as monitoring, management, optimization, and maintenance (Cimino et al. 2019). Furthermore, it can enhance process efficiency to increase the productivity and reduce the energy consumption, as well as predict future resource requirements (Park et al. 2020). The technology provides the capabilities of tracing product fault sources and analyzing production efficient bottlenecks (Lu et al. 2020c).

In the health care area, the digital representation of an individual, including personal data like health data, activity data, and emotional status, can assist in comprehending personal well-being and working conditions in a factory. This can lead to improving the psychological and physical health of employees and achieving upskilling and the best production performance through ultrarealistic training programs (Kendzierskyj et al. 2019; Lu et al. 2020c). In the aviation sphere, digital twin technology contributes to moving the aerospace industry from reactive to proactive maintenance based on predicting the structural behavior and asset failure by simulating an aircraft's behavior and pinpointing critical weak spots (Glaessgen and Stargel 2012; Lu et al. 2020c). In the agricultural domain, the digital twin can be utilized to predict climate change, monitor the growth of plants, and generate advice to farmers for ensuring harvesting at the most cost-efficient time. It also informs about changes in the microenvironment such as temperature, pressure, pH, and soil nutrients to provide improved feedback (Barnabas and Raj 2020).

Digital Twins' Application in Construction-Related Scenarios

The construction-related field has been exploited for smart cities, built environments, facilities, infrastructure, construction, properties or estates, workspaces, transportation, and parking systems. More specifically, in the smart cities scenario, digital twin technology can capture the special and temporal implications to optimize the city. It can accomplish this by detecting urban events, public traffic, and public environmental monitoring (Wang et al. 2019). Further, Francisco et al. (2020) identified that it can monitor and solve city environmental and energy problems to help realize low carbon intensity.

The built-environment domain utilizes digital twins to analyze climate, carbon, and circular economy outcomes (Kaewunruen et al. 2019), as well as energy use, air pollution, and water quality (Ham and Kim 2020). Fan et al. (2020) developed it to help in monitoring and analyzing the various situations during built-environment disruptions in disasters. In extreme weather conditions, based on the visibility of the disaster and the city interaction, decision makers can monitor and analyze various situations in the built-environment disaster and the impact on people (Fan et al. 2020; Ham and Kim 2020). Furthermore, it can improve the ability to deal with the destruction of the built environment in a disaster and community disaster management for community members, volunteer responders, residents, and other stakeholders (Fan et al. 2020; Ford and Wolf 2020).

In the property/estate sphere, Barnabas and Raj (2020) applied digital twins to analyze the estate plans to maximize the utilization of the space. In the workspace field, the digital twin can support optimal design and usage of office space by leveraging its occupants' health and performance data. This can enable proprietors

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to control the workspace better and enhance tenant satisfaction (Arup 2019). Furthermore, it can help in monitoring different sets of data from sensors in regard to occupancy, temperature, carbon dioxide level, and meeting rooms' status (Arup 2019).

For buildings, the most significant application aspect is energy performance assessment and management. One such instance is building energy analysis during the design phase (Lydon et al. 2019), and near-real-time energy management during the operation phase. Based on clarifying the annual energy consumption benchmark, it is critical to develop the daily building energy benchmark strategically. Combined with the use of digital twin technology to quantify the variation between daily energy performance and the benchmark target for adjusting the building energy plan retrofit strategy, it can be based on the impact indicators to finally achieve near-real-time energy management (Francisco et al. 2020). Lu et al. (2020a) developed a semiautomatic geometric digital twin method based on pictures and computer-aided design (CAD) drawings for existing buildings in the operation and maintenance stages. It has the ability to effectively access the latest information on existing buildings in the operation and maintenance phases, which helps to avoid decision delays caused by incorrect or outdated building information. Overall, it avoids the negative effects of the response to residents' daily requests and even emergencies. Therefore, this method can improve organizational efficiency.

There are few relevant literature studies the construction field. One specific study undertaken by Boje et al. (2020) put forward the transition path from the construction BIM tool to the construction digital twin based on semantics, after reviewing the multifaceted applications of BIM during the construction stage. This study indicated some potential methods and technologies to be considered for digital twins in the construction sphere. The study only touched on a few points concerning the construction site. It also summarized the limits and requirements for future construction digital twin development. Lin et al. (2019) proposed analogous digital twin–based structural safety management in bridge construction. The construction process, as well as the corresponding structural performance during construction, can be visualized through the integration of the structural analysis model and the simulation results, as well as the monitoring data.

In regard to infrastructure, it can be used to monitor infrastructure conditions (Fan et al. 2020) that contribute to decision-making for infrastructure risk management (Ham and Kim 2020). As for facilities, the digital twin technology can monitor the working status of a facility and further discover any anomalies in time (Lu et al. 2020b). In the transport systems area, the technology contributes to self-driving technology by robustly enhancing radar and image recognition as well as vehicle-to-vehicle communication capabilities. For instance, it can enable the prediction of bus arrival times, cloud-based driving direction systems, and active accident-avoidance applications (Wang et al. 2019). For city parking networks, the technology can identify underused locations to drive decision making on infrastructure reallocation (Arup 2019).

In summary, digital twin technology is currently applied in a multitude of fields, including construction. Although there are studies available on how to transition from BIM to digital twins in the field of construction, the papers only touched on a few points concerning the construction site. Few studies emphasized the specific aspects of digital twin technology for the construction process, such as the structural safety management of bridge construction. Nevertheless, there is a lack of research on the digital twin technology for completed construction site process management and its comprehensive application for construction sites.

LoDs for BIM

Many countries have formulated their own LoDs standards for the BIM domain. Table 1 summarizes some predominant LoDs standards related to construction, demonstrating the main classification principles with detail explanations. LoDs is interpreted as the level of detail or level of development in different standards.

Table 1. Existing predominant standards about LoDs of BIM

Area	Description	Standard
Hong Kong, China	Demonstrate the LoDs of site model. Especially present the LoDs of the site topography, natural and artificial slope, flexible and rigid barrier, massing models of adjacent or surrounding buildings, geological model, pavement, profile barrier, and so on [CIC (2019c)]. This standard highlights the LoDs of the site model rather than the construction site model.	• *CIC Building Information Modeling Standards General* [CIC (2019d)] • *CIC Production of BIM Object Guide General Requirements* [CIC (2019c)] • *CIC BIM Standards for Underground Utilities* [CIC (2019b)] • *CIC BIM Standards for Mechanical Electrical and Plumbing* [CIC (2019a)]
Mainland, China	Classify BIM into Deep level-G 1, 2, 3, 4, and 5 and Deep level-N G 1, 2, 3, 4, and 5 [Beijing Engineering Exploration and Design Association (2013)].	• Building Information Model (BIM) Design Basic Standards [Beijing Engineering Exploration and Design Association (2013)]
USA	Combine the levels of development 100, 200, 300, 400, and 500 with the design phase from preschematic, schematic, design development, construction documents, and services during construction [Bloomberg et al. (2012)]. Use 100, 200, 300, 350, and 400 levels, with a series of samples to explain the levels [BIMForum (2013)].	• BIM guidelines (New York) [Bloomberg et al. (2012)] • *Level of Development Specification For Building Information Models* [BIMForum (2013)] • *Level of Development Specification Part I & Commentary For Building Information Models and Data* [BIMForum (2019)] • *National BIM Standard-United States* [NBIMS-US (2015)]
Singapore	Separate the development into 1, 2, 3, 4, 5, and 6 levels, and combine the whole construction process into project brief development, concept design development, detailed design development, preconstruction, construction, and postconstruction [BCA (2016)].	• *BIM Essential Guide* [BCA (2016)]
UK	Divide the BIM maturity levels extended to asset information management into Levels 0, 1, 2, and 3 from CAD to 2D/3D, BIMs, and iBIM [BSI (2014)]	• PAS 1192-2:2013 [BSI (2013)] • PAS 1192-3:2014 [BSI (2014)] • PAS 1192-5:2015 [BSI (2015)] • PAS 1192-6:2018 [BSI (2018)]

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Level of detail is essentially how much detail is included in the model element. Level of development is the degree to which the element's geometry and attached information has been considered (BIMForum 2013). In addition, Trani et al. (2015) explored the LoDs for construction site design, classifying the levels into preliminary, clear, and execution levels and demonstrated that the detail level clarifies all design issues between different designers. However, that study focused on the LoDs in construction site design rather than the construction site management. Cassano and Trani (2017) explored the LoDs for construction site elements. They divided the levels into conceptual, developed, detailed, and installed. Their study considered the stages between the draft design and the detailed design, and the construction stage developed by contractors for the installation design. However, they did not consider the whole construction site life cycle.

In general, the major current gaps are as follows: (1) there is no existing research that considers digital twins for whole construction site process management and its comprehensive application for construction sites; and (2) no standards and literature are available that define the LoDs of BIM for construction sites management and its intuitive model example.

Proposed Digital Twin Framework

The digital twin presents the bidirectional dynamic mapping between a physical object and its digital model in cyberspace (Tao et al. 2018). The establishment of a digital twin framework for construction site management requires four core components working together:

1. A physical environment.
2. A digital environment containing the digital model that abstracts the characteristics of the physical counterpart and a data processing module that is capable of extracting information from heterogeneous multisource data to construct real-time representation of physical objects (Lu et al. 2020b) that facilitate the flow of information from raw data to high-level comprehension for supporting humans decision making and further functions at different levels (Liu et al. 2018). Standards play critical roles in providing information models that describe various physical objects (Lu et al. 2020c).
3. A communication mechanism used for transmitting two-way data between the digital and physical twins with real-time data obtained through IoT state synchronization (Lu et al. 2020c).
4. Feedback to the physical environment application.

Fig. 1 illustrates an overview concept map of the four core components' work machine within the proposed digital twin framework for construction sites management.

Components of Digital Twins

Digital Representation for Construction Sites

The NBIMS-US (2015) states that BIM is a technology that is able to upgrade the whole life cycle of the construction industry from design, construction, operation, and maintenance (Panushev et al. 2010). It establishes corresponding information models for building components and equipment components, among others, and collects information corresponding to components. Three-dimensional modeling of BIM for specific construction projects can provide stakeholders with a visualization function to show the overall situation of the project and its components (Davis 2003). Furthermore, BIM allows centralized storage of information, rapid updates, and multistakeholder access to reduce file waste by reducing task redundancy, thereby achieving database efficiency. BIM is a platform that enhances collaboration, efficacy, and quality.

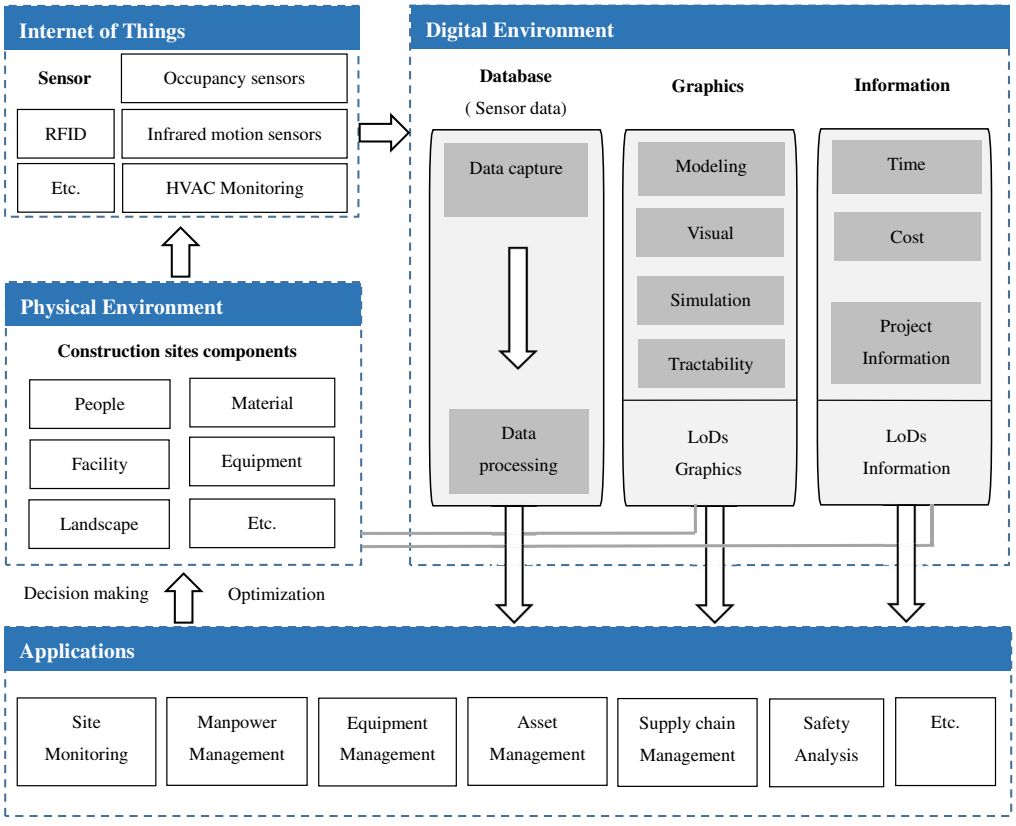


Fig. 1. Concept map of digital twin for construction sites.

J. Manage. Eng., 2022, 38(2): 04021094

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It enables users to integrate and reuse information throughout the project life cycle (Ozturk 2020). BIM provides consistent, coordinated, and computable information management within a sophisticated and fragment construction project to make information and knowledge retrievable and traceable (Becerik-Gerber et al. 2012). This significantly improves the computing and data capabilities of the construction industry.

To explain the application scenarios in the construction site process, this research considers the construction site process from inception feasibility, design authority, tender, procurement, and preconstruction to construction stages. Concerning the initial feasibility, historical site usage data are collected to deploy analysis in the digital model for determining the site condition. For the design stage, the digital model provides an intuitive and visual design representation to facilitate cross-professional communication and collaboration, and jointly optimizes the layout design plan to meet the actual requirements (Ning et al. 2011) and multiple objectives. In the tender and procurement stages, time and cost, as well as other data, are added to reflect the project economy and schedule planning visually. In the preconstruction stage, physical conditions such as the weather data around the site, as well as channels, transportation, and logistics data and so on can be fed back into the model for deducing the required simulation to predict future problems, and then optimize the solution. For the construction phase, adding real-time data contributes to monitoring the construction site in real-time and optimizing the usage of facilities and equipment, thereby saving money and adapting to the actual use of the construction site. Furthermore, it can be utilized to synchronize the plan, check collision conflicts such as the complex integrated underground pipelines, distance measurement, automatic manufacturing, quantity measurement, and the estimation of component equipment, among others.

Internet of Things

The Internet of Things is an extended and expanded network based on the Internet (Miorandi et al. 2012). Through sensing devices, all items are connected to the Internet through the agreed protocol and open standards to achieve intelligent identification and management (Sheng et al. 2013). The vital sensing hardware infrastructure comprises radio-frequency identification (RFID), near-field communication (NFC), and sensor networks (Whitmore et al. 2015). The sensing devices can be utilized for different construction site components to achieve certain functions. The main components of the construction site include site space, boundary, road, facility, equipment, material, and people, among others.

Regarding the site space, such as the working area and living area, turning on the lights in a place without people wastes light energy, except for special situations in which lights are needed on the construction site. The use of occupancy sensors and light sensors helps to automatically control and adjust the light system, as well as manage energy consumption on the construction site (Garg and Bansal 2000). For managing the construction site boundaries, sensors can be installed to detect when a door is opened or closed for identifying any unauthorized access. Utilizing sensors and actuators to track energy consumption in the construction site, moreover, allows monitoring and controlling such building infrastructure as HVAC systems. Construction site safety requirements can be satisfied by conducting routine surveillance.

Concerning equipment such as construction tower cranes, encoder sensors are installed to detect the horizontal angle of a tower crane by quickly calculating the number of spins or the plumb for preventing collisions (Lee et al. 2012). It can also support equipment safety management by tracking equipment movement and predicting operator blind spots (Vasenev et al. 2014). In terms of personnel management, appropriately distributed wireless sensors on work clothing, combined with applications, can be used to locate a person and identify whether they are close to or in a hazardous area (El Amine and Mohamed 2013) of the construction site, and then verify the required security measures. Furthermore, personal health and work status can be understood by measuring the heart rate and body temperature (Stankovic 2014) to prevent personnel accident occurrence during operation on the construction site. The IoT can provide detailed, visual, and real-time information of the supply chain and logistics and further achieve automatic tracking and management, such as for adjusting the conveyance route and the number of trucks that transport construction material in order to improve the efficiency in the process of transportation (Vasenev et al. 2014). It can also be applied to trace construction materials for managing quality and safety.

Data Storage, Integration, and Analytics

The data processing module is one of the core components of a digital twin system. It can extract information from heterogeneous multisource data and realize high-level comprehension from the raw data through a series of processing stages (Lu et al. 2020c). This section divides data processing into three phases: acquisition, integration, and usage. It indicates the data processing from data capture to usage in the digital twin, and points out the potential problems that need to be solved in each stage, as well as possible solutions and suggestions. Fig. 2 illustrates the data flow from raw data to high-level understanding after the three mentioned processing phases.

The data capture mechanism from physical devices utilized in digital twins generally employs two approaches, namely, capturing

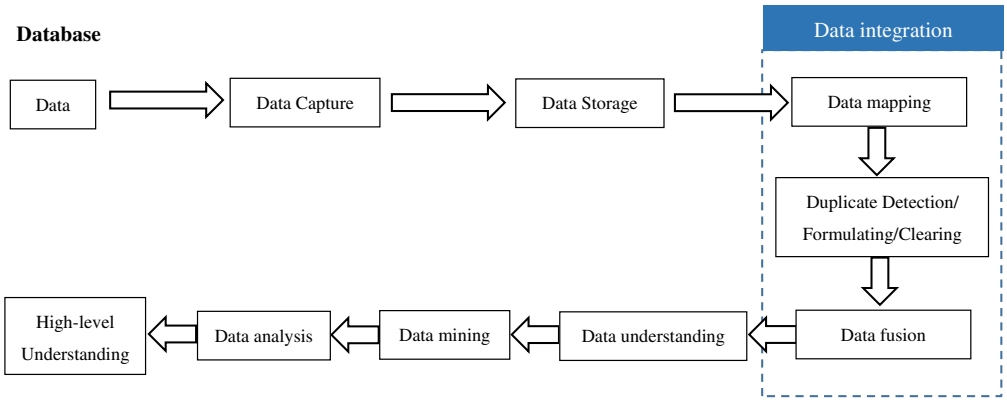


Fig. 2. Data processing map from raw data to high-level understanding.

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changes and taking snapshots (Lu et al. 2020c). For data storage, time sensitivity is the key issue in achieving low latency (Lu et al. 2020c). The utilization of digital twins effectively facilitates real-time monitoring, controlling, and prediction, and the identification of potential issues with the real physical counterparts (Lu et al. 2020c). Accordingly, digital twins have high requirements for the timeliness of data processing.

Data integration mainly covers three stages, namely, data mapping and data duplication detection, as well as data fusion. Generally, the problem that existing data management tools cannot solve is that information sources do not share common terminology. In particular, different data formats and architectures, as well as various analysis tools and standards, should be considered. The main goals of data integration are to improve the integrity by adding more data sources, describing the characteristics of the attributes, and enhancing the simplicity of the data available to users and applications by eliminating redundant content (Bleiholder and Naumann 2009). Data mapping is the first step in data integration, which refers to the process of establishing the correspondence between different data models using a set of definitions.

The second step is duplicate detection. The goal of this step is to identify multiple semantic representations of the same real-world entities (Ramya et al. 2017). Data fusion is the key technology driving digital twins. The process of data fusion fuses several records corresponding to an identical real-world entity into a single, consistent, and clean tuple that represents the joint information to a maximal extent (Bleiholder and Naumann 2009; Bronselaer et al. 2014). It provides specific benefits such as better signal quality, model performance optimization, prediction improvement, and a more reliable basis for decision-making (Liu et al. 2018).

The third step in data processing is to achieve a high-level understanding of the raw data (Kaur et al. 2020), which is driven by the information fusion technologies. There are some applications in construction sites. For example, by combining the workload data generated by workers in the worker-power management platform and the construction site sensor data, a dynamic model can be built. The decision support mechanism consists of prediction algorithms that support more insights through data fusion (Wang et al. 2019) for more accurately arranging daily worker power and worker types.

Interaction with Physical Environment

The digital twin seamlessly merges the practical and virtual worlds through continuous data exchange. The change in the practical world directly affects the virtual world and vice versa. Through integrated computing, the entire process of the digital and physical worlds is defined as closed-loop (Guan et al. 2016), where embedded computers and systems usually control and monitor physical processes. In return, the physical processes can affect computing. It should be emphasized that integration does not mean a simple fusion of the physical world and digital world, but that there is profound interaction between the physical and digital components. This interaction can be divided into two types: one is machine-to-machine interaction, and the other is human/object-to-machine interaction.

For machine-to-machine interaction, the digital twin is able to achieve dynamic reconfiguration and reorganizing with a high degree of automation in multiple temporal and spatial dimensions due to the tight coupling and coordination between the digital and physical worlds (Ding et al. 2018). The collected data are processed and automatically analyzed, and subsequently send feedback to the monitoring, controlling, and management loop in the system. Therefore, on the one hand, problems such as model deviation between the digital and physical worlds can be automatically adjusted and corrected, the quality of the construction site model can be improved. On the other hand, it is possible to respond flexibly to disruptions or modified requirements (Wang et al. 2015).

In terms of human/object-to-machine interaction, the digital model feeds back the monitoring results, simulation results (Boschert and Rosen 2016), analysis support, and so on, to the physical construction sites, contributing to a series of construction site process management actions such as worker-power management, equipment management, asset management, supply chain management, safety analysis, and emissions and energy management, among others. The digital twin can provide support for different stages of the life cycle in the construction site. The digital twin provides an intelligent scene of information on the available systems to enable participants in the various disciplines access to the models and results at an early stage, and then optimize the plan for the following stage. The digital twin model created in the process of design can be reused in a subsequent process, and the information it contains can interact with actual data using appropriate interfaces (Guan et al. 2016).

The actual data can be used to evaluate and verify the simulation model during a later process operation, as well as the input and output of new information, which enables continuous engineering improvement. For example, if the project's expectations have been changed and need to be modified, such as in the case of collision problems found on the site, and time schedule adjustment problems, among others, the existing model can be utilized and modified to output the adjusted construction guidance plan. On the other hand, if the operation data is also collected systematically as part of the digital twin, it can be used to verify and update the existing model under actual operating conditions, so that the acquired knowledge can also be used for similar engineering in the future. The contribution of the interaction with the physical environment in construction sites application can be explained as follows:

- Site monitoring: The digital twin can provide users with an interface that displays the information and real-time status of all the systems for promoting more intuitive data analysis for the construction site manager. The manager can view the different aspects that occur on the construction site and at the same time, extract various data bits and make visual comparisons and associations, and then make accurate, reasonable, and beneficial decisions. Notably, in the face of emergencies, the manager can find problems more quickly and respond more efficiently.
- Equipment management: An interface that visually displays the equipment status can provide maintenance or replacement early warnings and visually display the fault location in the digital model to prevent equipment damage from affecting the construction plan and delaying the construction process. Equipment usage records, repair records, maintenance arrangements, and scrap plans can be retrieved from the platform whenever necessary and contribute to the lease and purchase adjustment and arrangements.
- Asset management: Asset-related data can be directly displayed in front of the manager through data visualization technology to support decision making.
- Supply chain management: Supply chain route simulation and its rationality and limitations analysis can be used for route adjustment, logistics equipment, and personnel arrangements.
- Worker-power management: Workload data contribute to managers' accurately arranging daily worker power and worker types and optimizing adjustment programs based on labor cost calculations. The corresponding number of scaffold rentals may also be adjusted.
- Safety analysis: On the one hand, in the construction site design process, a reasonable structure must be designed so that it can

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bear a given load for safety guarantees. Sensors are installed to monitor operations and provide fault warning and structural health monitoring (Hackmann et al. 2013). Combined with the digital model simulation, several failure modes can be simulated instantly in an attempt to reproduce the actual measurement signal (Hackmann et al. 2013). Furthermore, comparison of the simulated and measured signals help identifies failure modes. On the other hand, the health status and location of personnel can be monitored for identifying whether they are approaching dangerous areas on the construction site and preventing personnel accidents occurrence during operation at the construction site.

- Emissions and energy management: The digital twin is able to collect the construction site carbon emissions and energy performance data through sensing devices, then simulate and predict the emission sand energy situation to provide a basis for the consumption and energy efficiency optimization strategy.

LoDs of Digital Twins for Construction Sites

In this section, the classification for the components in the construction site is first illustrated for supporting the description of the LoD division. A hierarchical insight of the construction site components is given through the tree map (Fig. 3). The second column of the subbranch is classified according to different component categories. The third column is classified according to different functions of the same category. The fourth column signifies detailed elements. Generally, the elements in the construction site include the site space, such as the working area, living area and business area, boundary, road, landscape, facility, equipment, and prefabrication, as well as material and people. In addition, the LoD-Graphics for construction sites is proposed from an overall point of view, pointing out the principles and the required components in the different subdivision stages.

The aforementioned are followed by the LoD-Graphics for construction site components that describe the detailed division of separated components under the same division principle that refer to approximate, basic, detailed, completed, and verified levels. The minimum LoD requirement for construction site components directly explains the LoD relationship between the LoD-Graphics for construction sites and the construction site components. To be more specific, it illustrates what components and what LoDs of the components should be included in the LoD 300. Subsequently, the LoD-Information requirement is presented that contains the project information, time data, and cost data.

LoD-Graphics for Construction Sites

Based on the LoDs, division principles, from approximate to basic, are detailed, completed, and verified. The five LoD-Graphics for construction sites are summarized as follows:

- LoD 100: the model elements are graphically represented within the model by a rough three-dimensional (3D) shape.
- LoD 200: the basic space in the construction site is clearly divided and the basic framework of the construction site is graphically indicated.
- LoD 300: the construction site detailed plan and preparation can be clearly identified.
- LoD 400: the construction site formation should be represented as a complete and accurate 3D model.
- LoD 500: the model element is a field-verified representation.

Further interpretations and comparisons among these different standards are given next, and supporting examples and more detailed explanation are shown in Fig. 4. All information obtained from LoD 100 elements must be considered approximate. In comparison with LoD 100, the space division on the site is evidently reflected in LoD 200, including a number of allocated spaces for operation and maintenance. In comparison with LoD 200, specific mounting details are included in LoD 300. In contrast with LoD 300, the model components are more detailed and independent in LoD 400. Any information is derivable from this level graphic model, such as time and cost. More specifically, the BIM four-dimensional (4D) model presents a project time process and the association among components. Here, 4D time corresponds to the order and work schedule. Therefore, the details should comprise the construction order information.

The BIM five-dimensional (5D) module should include a database system to collect, analyze, and utilize related statistics like area, volume, unit cost, and total price, among others, as well as generate reports to provide information regarding project management decisions. Accordingly, in 5D, each component in the digital model must be clear and independent. Each detail should be illustrated independently; otherwise, the manager is prone to make errors when calculating the areas through the model, which would inflict a difference on the final cost of the materials, thereby affecting the accuracy of the assessment. For instance, all layers of the panels in wall components require calculation to get the accurate cost. A similar scenario applies to the specific diameter and length of each pipe in a network of underground pipes and the exact number of fire sprinklers; all of these relate directly to the cost.

LoD-Graphics for construction site components describes the detailed division of the separated components. The summary and examples are demonstrated in Fig. 5. It should be explained that the stage of appearance of some components may not be from the beginning, such as equipment and landscape, which appear in relatively late stages.

The LoDs provide a snapshot of the elements from concept to specified process. The AIA pointed out that only when all the requirements specified in the definition are met can the element develop to a given LoD (BIMForum 2013); as a result, the concept of minimum requirements for LoD is proposed. It should be emphasized that demand is cumulative. For a given element, each LoD's definition includes all previous LoD requirements. For example, to make an element qualify for the conditions of LoD 300, it must meet all the requirements of 200 and 100 and the requirements specified in the definition of LoD 300 (BIMForum 2013). Table 2 indicates the LoD relationship between the LoD-Graphics for construction sites and construction site components.

LoD-Information for Construction Sites

The main information required in a construction site is project information, time (4D), and cost (5D). Based on the same division principles as LoD-Graphics, the information from the 100 to 500 detail levels are developed from approximate to basic, detailed, completed and verified. The LoD-Information requirements are given in Table 3.

The construction site project information types considered are

- site space: size, location, and project information;
- roads: width, length, level, and position;
- equipment: number of facilities, type, and models;
- materials: weight, volume, distribution, and delivery time; and
- people: training record, age, gender, different work types, and number and work types in different stages.

The generated cost data are based on the budget and final accounts associated with various stages of a project. The budget quota of the construction project considers the preliminary design, cost of equipment and materials, and other related costs. At the construction scale, the structural nature and type of equipment are modified and altered, and the preliminary design changed; hence, the preliminary estimate is adjusted accordingly. In other words, based on

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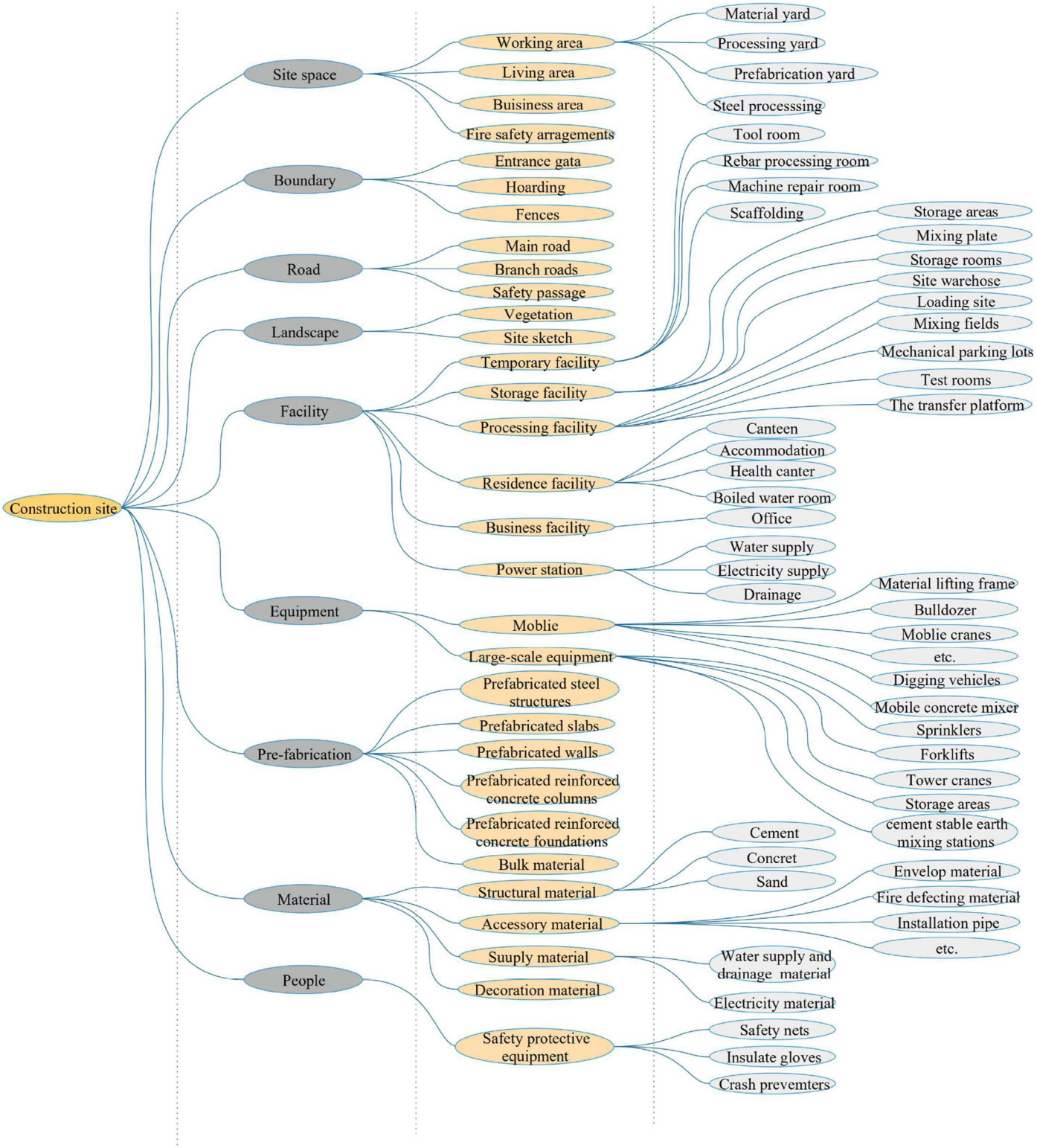


Fig. 3. Components in the construction sites.

the available data, numerous methods are used to predict the revised budget estimate for the future of the construction project; this process is known as investment estimation. The liquidation price of the complete projects is calculated by the contractor in accordance with the contract agreement. The finalized accounts of completion reflect the actual costs and investment effect of the project, including a comprehensive reflection of the economic effect of the capital, as well as demonstrating the foundation of the project delivery and its use.

BIM Uses for Construction Sites

This section considers the construction site process from inception feasibility, design authority, tender, procurement, and preconstruction (planning) to construction stages. BIM can support a number

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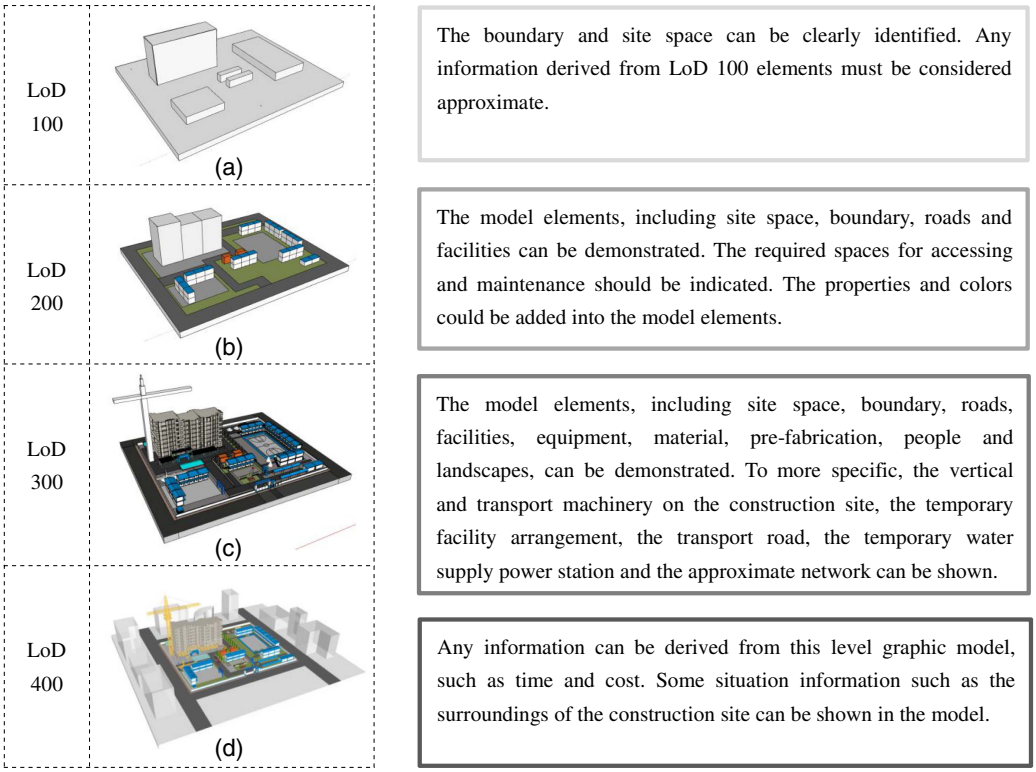


Fig. 4. Details and examples of LoD-Graphics for building construction sites.

of detailed actions during construction site processing. The following explains the definition and requirement of each detailed action based on *Construction Industry Council (CIC) Building Information Modelling Standards General* (CIC 2019d). The minimum LoD supported for different uses at different stages described in this section is a combination of LoD-G and LoD-I, as presented in Table 4. If not specifically highlighted, LoD-I uses the same level as LoD-G:

- The minimum level of morphology-related details supported is 100, including in existing site modeling.
- The LoD minimum support related to space division is 200 and includes (1) site analysis, which refers to use BIM and Geographic Information System (GIS) tools to assess a site and determine the optimal location, position, and orientation for a future project; (2) space programming utilizes a spatial program to assess a design layout, space definition, and detailed space zoning in terms of spatial requirements; (3) phase planning (4D modeling) aims to demonstrate the construction sequence and space requirements in a construction site; and (4) site utilization planning, which is related to the construction schedule (4D) and is used to examine space planning, site logistics, sequencing requirements, temporary works, and safety.
- The minimum LoD support for analysis, collaboration, and control is 300, which involves (1) 3D coordination needed to avoid collisions and to decide the criteria for headroom and the areas designated for work before the commencement of a construction project; (2) cost estimation; (3) engineering analysis; (4) facility energy analysis; (5) 3D control and planning, which refers to the digital layout of certain project elements such as wall positions; (6) project system analysis, which include airflow, solar gain, and lighting analysis; (7) maintenance scheduling (The LoD-I support for this stage is 400 due to the detailed scheduling requirement); (8) site space management and tracking, which refers to the as-built model that can be employed in a particular project to assess, manage, and monitor spaces and related resources; (9) asset management (The LoD-I support for this stage is 400 because of the specific details requirement); and (10) site monitoring.
- The minimum support for the LoDs relevant to the as-build model and that requiring special details is 400, including (1) cost control; and (2) digital fabrication, which means the utilization of models to enable fabrication of construction materials or assemblies.

Data/Information Management

For a construction site, data management issues mainly involve multiparty data management, project information management, professional technical information, data cooperation and sharing, and security and privacy issues.

The construction process has a fragmented and complex structure. It is an information-intensive process because a large volume of information is exchanged and transferred during the project life cycle (Chen and Kamara 2011). The traditional paper-based information management procedure leads to reworking, inefficiency, data loss, and falsification (Chen and Kamara 2011). The BIM model provides the possibility of solving these problems. BIM can cover the entire stage of the project process, from start to delivery, from which the information flow of the building information model in the whole life cycle is unified and open, the interaction process is efficient, and the model is digital and visual (Ma and Ma 2017). Information standardization and integration are tools for data loss prevention and defragmentation, resulting in better information sharing, usage and reusage, as well as all parties being involved through communication coordination (Ozturk 2020).

The transmission and sharing of project information resources are relatively difficult to achieve in the traditional approach,

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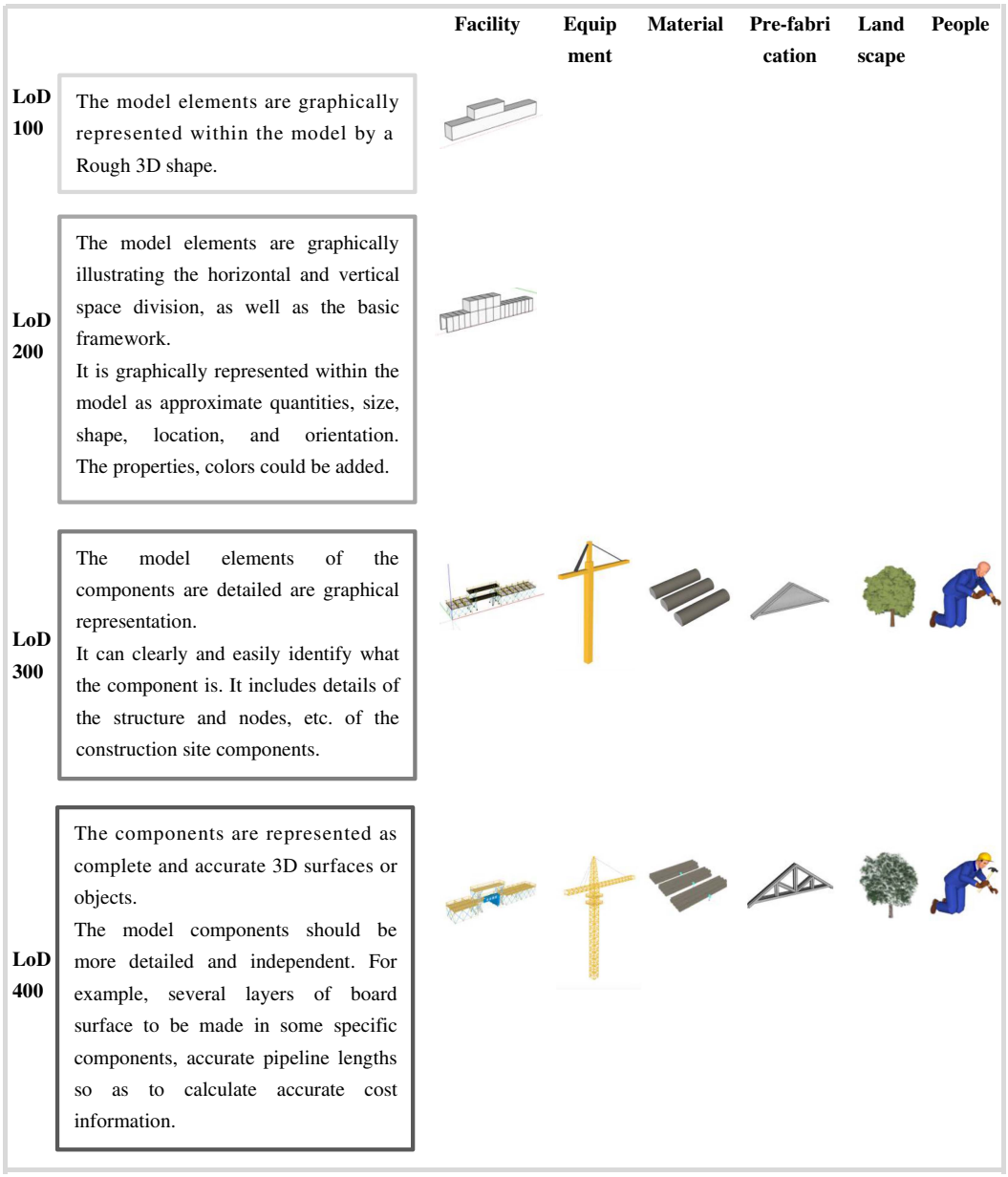


Fig. 5. Summary and examples of LoD-Graphics for construction site components.

Table 2. Minimum LoDs requirement for construction sites components

	LoD 100		LoD 200		LoD 300		LoD 400	
Component	G	I	G	I	G	I	G	I
Site space	100	100	200	200	300	300	400	400
Boundary	100	100	200	200	300	300	400	400
Road	—	—	100	100	300	300	400	400
Facility	—	—	100	100	300	300	400	400
Equipment	—	—	—	—	300	300	400	400
Material	—	—	—	—	300	300	400	400
Prefabrication	—	—	—	—	300	300	400	400
Landscapes	—	—	—	—	300	300	400	400
People	—	—	—	—	300	300	400	400

Note: G = graphics; and I = information.

resulting from the influence of the transformation of each stage and the corresponding organizational relationships. There are information islands and fragments, as well as data duplication between different stages of the life cycle (Tao et al. 2018). In addressing this, if the information generated during the entire life cycle is stored and integrated into a data model, the information can continue to flow and be shared throughout the life cycle.

The collaborative operation of different professional backgrounds in the information model needs to integrate all the professional knowledge and technical data involved in the project. However, it is still a challenge to incorporate all the expertise and technology required in the project into the building information model nowadays.

The achievement of the project target is based on the cooperation of the project stakeholders, which requires the full range of the relevant skills and professional knowledge of the project participants, and the sharing of information and resources. However, in the specific construction site process, in regard to sensitive private information, who should be given the authority for particular information is also a question to consider in terms of, on the one hand, preventing waste of resources, and on the other hand, avoiding the risk of personnel turnover in the project.

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Table 3. LoD-Information requirement

LoD	Project information	Time (4D)	Cost (5D)
100	Approximate information	Project start time and end time, total working days	Budget quota
200	Basic information	Time-schedule is based on segmented target/component, phase time estimates	Revised budget estimates
300	Detailed information	Processing order, time node and bridge node for different stage/component, equipment installation entry time, specific facilities when to rent and exit equipment use time arrangement data, inventory data guide procurement	Investment estimation
400	Completed information	Project real-time statistics	Liquidation price of the completed project
500	Verified information	Planning and actual construction progress comparison, plan and actual installation progress comparison, valid working days	Final accounts of completion of the actual cost documents and investment effect of the project

Table 4. Explanation of the minimum LoDs supported for different uses in different stages

		LoD-G				LoD-I			
Stage	BIM uses	100	200	300	400	100	200	300	400
Inception feasibility	Existing conditions modeling	Yes	—	—	—	Yes	—	—	—
	Site analysis	—	Yes	—	—	—	Yes	—	—
Design authoring	3D coordination	—	Yes	—	—	—	Yes	—	—
Tendering	Cost estimation	—	—	Yes	—	—	—	Yes	—
Procurement	Space programming	—	Yes	—	—	—	Yes	—	—
	Engineering analysis	—	—	Yes	—	—	—	Yes	—
	Facility energy analysis	—	—	Yes		—	—	Yes	—
	Cost control	—	—	—	Yes	—	—	—	Yes
Preconstruction (planning)	Phase planning (4D modeling)	—	Yes	—	—	—	Yes	—	—
	Site utilization planning	—	Yes	—	—	—	Yes	—	—
	3D control and planning	—	—	Yes	—	—	—	Yes	—
	Digital fabrication	—	—	—	Yes	—	—	—	Yes
Construction	Project systems analysis	—	—	Yes	—	—	—	Yes	—
	Site monitoring	—	—	Yes	—	—	—	Yes	—
	Site space management and tracking	—	—	Yes	—	—	—	Yes	—
	Asset management	—	—	Yes	—	—	—	—	Yes
	Maintenance scheduling	—	—	Yes	—	—	—	—	Yes

Validation

This study adopted semistructured interviews and questionnaire analysis to test and validate the proposed framework. Questionnaires were conducted to confirm the significance of the proposed framework's contribution and to check whether the components included in the framework were crucial. Interviews were conducted to review the framework and its component associations, application coverage, and corresponding LoD definitions, and to obtain in-depth comments and to determine the problems encountered in the practical application of the framework. After comprehensive consideration, 31 questionnaires and 10 interviews were conducted from July 21 to August 6, 2020.

In the questionnaire survey, there were 31 participants, and all the samples were valid. This number was considered sufficient to balance the time constraints with the need for a robust sample. The years for which respondents have been engaged in construction ranged from 1 year to more than 20 years, and the distribution of the construction processes in which they were involved included inception feasibility, planning preliminary, scheme design, detailed design, authority, submission to approving, tender, procurement, preconstruction, construction, and research. Overall, the vast coverage and general participation make the results more objective and rigorous. The working area of the participants covered Mainland China, Hong Kong, China, the UK, the US, Dubai, Australia, and Singapore. The feedback from such multiregional participants made the validation of the framework more internationalized. The main questions in the questionnaire were as follows:

- Q1: The digital twin can support process monitoring on the construction site.
- Q2: The digital twin can support construction site management such as the manpower, equipment, asset and supply chain management, etc.
- Q3: The digital twin can play a role in improving the construction site quality.
- Q4: The digital twin can play a role in enhancing the construction site efficiency.
- Q5: The digital twin plays a role in improving the construction site safety.
- Q6: How can BIM benefit construction site management?
- Q7: How can IoTs benefit construction site management?
- Q8: How can the database benefit construction site management?
- Q9: Is the existing LoD's definition for construction sites clear?
- Q10: Is the existing LoD's requirement for construction sites clear?
- Q11: The LoDs extension facilitates the description of BIM uses expected in project plans.

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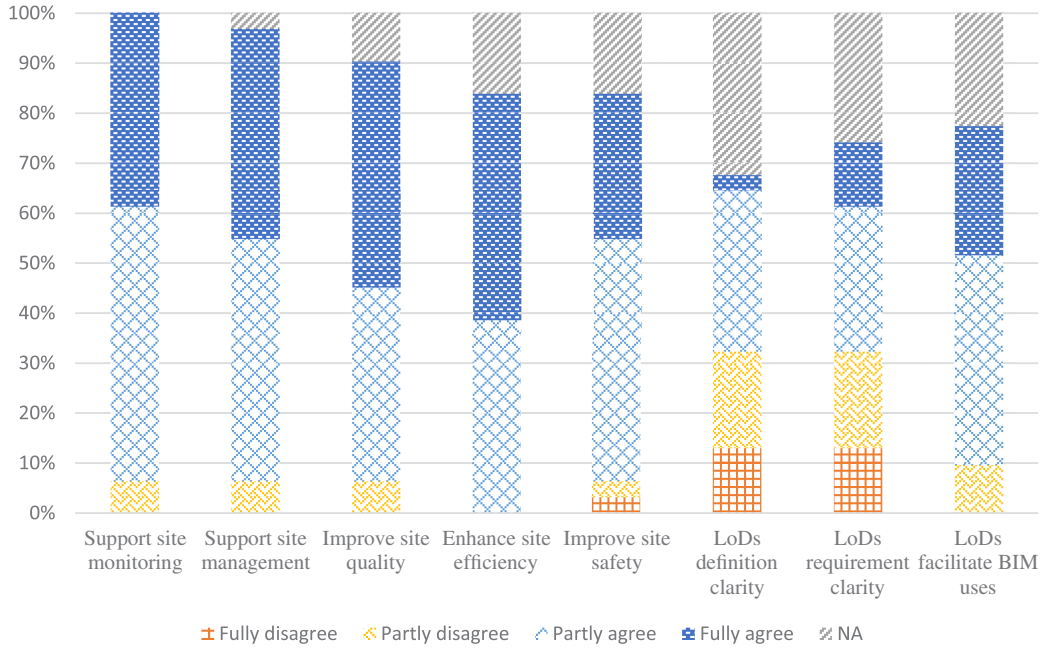


Fig. 6. Answers about the contribution of digital twin for construction sites and the LoD's meaning.

As shown in Fig. 6, an overwhelming number of respondents agreed that the digital twin can support process monitoring on the construction site and can support construction site management, such as in worker power, equipment, and asset and supply chain management, among others. It can play a role in improving the construction site's quality and efficiency. It should be emphasized that more than 77% of respondents agreed that the digital twin can play a role in improving construction safety. In addition, more than 68% of respondents agreed that the LoD's extension facilitates the description of BIM uses expected in project plans. Moreover, 32% of respondents thought that the existing LoD definitions for construction sites were partly clear, and 3% considered them fully clear. Additionally, 29% of respondents thought that the existing LoD requirements for construction sites were partly clear, and 13% considered them fully clear.

For further exploration, a short survey was conducted with the respondents who agreed that the LoDs for construction were clear, and they admitted that there is no national- or regional-level standards directly related to LoDs for construction sites, and that the company for which they work has customized LoDs standards related to construction sites, such as the China State Construction Engineering Group. As shown in Figs. 7–9, regarding how the BIM, IoT, and the database can benefit the management of construction sites, all benefits options have been selected although varying in degree. With respect to the questions regarding other benefits of BIM for construction site management, one participant supplemented "the approval of claims" in the questionnaire answers.

The interviewees' information and background are listed in Table 5. All of the interviewees were proficient in the parts flow of the framework, such as BIM and IoT for construction sites. Even some experienced managers have participated in almost the completed flows. Moreover, the interviewees had over 10 years of professional experience and had participated in multinational construction projects. In summarizing the feedback from the interviews, all of the interviewees agreed that the framework can positively impact on practical construction sites management.

Interviewees 1, 8, and 10 pointed out that the framework comprises features of strong logicality and a decent theoretical

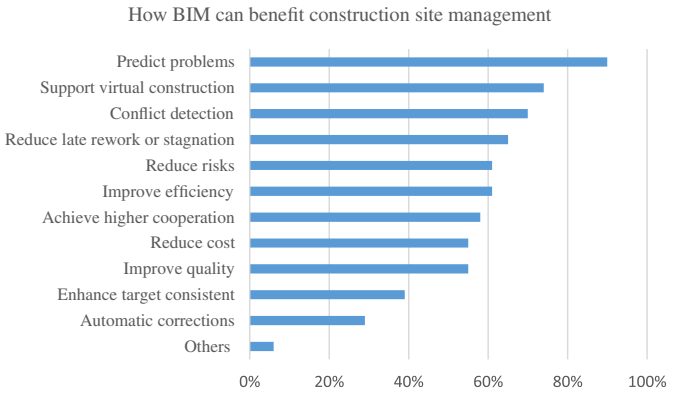


Fig. 7. Answers about the ways BIM benefits construction site management.

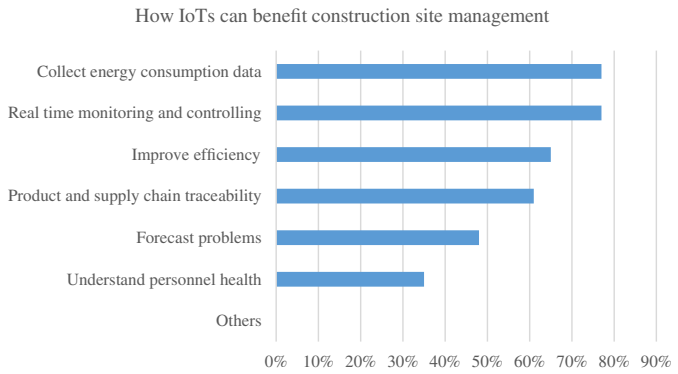


Fig. 8. Answers about the ways IoT benefits construction site management.

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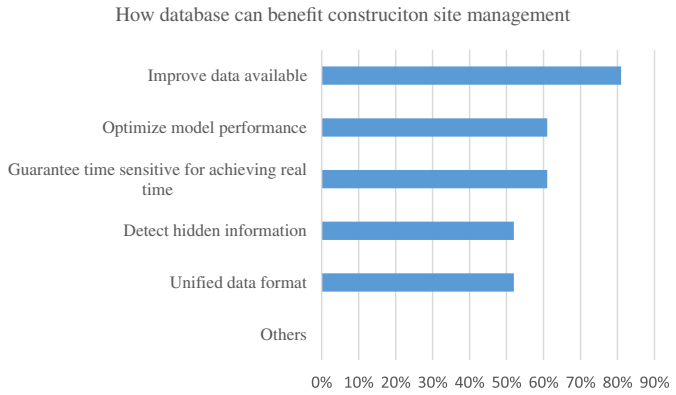


Fig. 9. Answers about the way of database benefit construction site management.

workflow. There were some findings to further validate and supplement the framework details. This framework has the capability to improve the safety and quality of the construction in many aspects (Interviewees 1, 3, and 9), such as analyzing the safety of the structures and measuring implemented prior to commencing onsite activities (Interviewee 1) as well as conducting quality control and technical inspection (Interviewee 10). Moreover, this framework can contribute to analysis of the staff health (Interviewee 9). Performing virtual training on the facility; aids in analysis of the overall behavioral of staff and their response to incidents (Interviewee 4). It also makes contribution to construction sites management improvement in a certain degree (Interviewee 7).

There are some potential problems that this framework can encounter in practical applications. Digital twins are a powerful tool, although how to maximize the benefits of this technology mainly depends on the users (Interviewee 1). Operators lack sufficient understanding of the technology and lack experience in the construction industry, which will in turn affect the accuracy of the inputted information and the outputted information. There is considerable uncertainty in the BIM information input process, mostly due to factors arising from human factors. If incorrect information has been added, then, when BIM export data are fed back to the physical world, there will be errors or inaccuracies. In fact, people with rich construction experience are unlikely to engage in basic modeling and data input processes (Interviewee 1), so training is important. Interviewee 4 held a similar view that the right people can help to achieve value. He has developed VR based training to analyze the understanding of a site team to tailor the correct training based on the staff behavior and response. The development and implementation of this system can support and resolve the conflicts mentioned before.

Considering cost cutting through BIM, Interviewee 2 clarified that along with these tools, checking and planning upfront will reduce costs, especially on reworking and efficiency. There would be initial costs; however, it will be much less of an increase when compared with the cost of reworking and/or efficiency. Regarding the IoT aspect, the instability of the IoT device leads to data loss and it is impossible to truly simulate the rea situation. Environment factors such as temperature, humidity and electromagnetic waves could interfere with the signal acquisition and transmission process (Interviewee 6). Construction in complex working conditions, such as in underground settings, would result in poor signal transmission. The high temperature and humidity in underground conditions will speed up corrosion, which will in turn affect the device (Interviewee 3). Generally, IoT technology service companies lack professional background in the construction industry. As a result, the solutions presented could lack a professional approach.

Referring to data storage, integration, and the analytics process, Interviewee 5 considers the biggest challenge is obtaining reliable data. Big Data processing primarily faces the problem of incomplete data and lack of structure because in the practical construction field, data collection and storage lack sufficient quality (Interviewee 3). Considering the data sensitivity and real-time problems, at the bandwidth of the telecom operator NB, it would be difficult to support large-flow signal transmission, such as images. With the high flow rate, data packets are often transmitted intermittently

Table 5. Interviewees' information and background

Interviewee	Position	Work location	Expertise areas
Interviewee 1	Architect/project manager, construction company	Hong Kong, China	Construction project management
Interviewee 2	• Head of project management, construction management and consulting company • Executive director, international construction consulting company	Indonesia China Malaysia	Construction project management
Interviewee 3	• Assistant dean, building information engineering technology research institute • Senior transportation planner, engineering consulting company	Canada China Singapore London	5G, IoT, and Big Data for urban infrastructure and transportation construction
Interviewee 4	BIM and engineering manager, construction contractor company	Dubai	BIM and construction project data integration
Interviewee 5	Head of research, international consulting company	US Asia China	Big Data and GPS for real estate and shopping center construction
Interviewee 6	Director, international consulting company	London India China	IoT and Big Data for residential and office building works
Interviewee 7	President, construction company	US China	Construction project management
Interviewee 8	Construction site manager, construction company	UK	Construction site management
Interviewee 9	Construction site manager, construction company	UK	Construction site management
Interviewee 10	Procurement manager, construction and project control management	Istanbul	Procurement for oil, gas, and petrochemical construction

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at a fixed time. One solution is using optical fiber sensors, which have unique advantages in continuous online monitoring/data acquisition (Interviewee 6).

Future Directions

At the current stage of the digital twin, there are both technical and nontechnical challenges. At the technical level, one of the goals of the digital twin in construction sites is to accumulate knowledge from design to construction, which helps to continuously improve the process and support optimization and decision making. The challenge is to establish a more comprehensive and dynamic digital model that integrates construction site life cycle monitoring, management, and operation, so that the entire process model can be constantly adjusted and updated according to the corresponding physical environment over time.

At the nontechnical level, existing digital twin applications are primarily used to monitor and predict purposes and as support for human decision making. Accordingly, human participation is essential in the digital twin–driven intelligent manufacturing environment. However, it is necessary to establish feedback and control from the digital twin to the physical world with a certain degree of autonomy. It is possible to complete many low-level operations autonomously without the human involvement, whereas various decision-making activities must be carried out by humans. The current challenge is that the human element is still not seen as part of intelligent manufacturing systems. In the future intelligent manufacturing environment, more research should be done on the importance and effect of human participation in digital twins.

Another major challenge is how to combine the nontechnical factors in construction projects to achieve the objectives. This is a potential point for future research considering the aspects of construction governance, such as transparency of information, suppressing project corruption effectively, and social benefits improvement, such as labor employment, and market benefits enhancement such as meeting market demand and promoting the development of upstream and downstream industries.

Additionally, ethical and moral data issues are challenges for digital twin databases, and they can be partly handled through hardware, data, and permissions. The hardware should conduct device management and security audits for IoT devices to enable the devices for security use. For data, on the one hand, permissions and authentication to clarify who the data are open to and shared with should be set up in order to prevent unauthorized use of data. On the other hand, identifying and intercepting fake and false sensor data, and then identifying the correct data and encrypting the data to protect sensitive data, is required.

Conclusions

Process integration and collaboration are considered as the key enabling technologies that drive the construction industry to increase productivity and efficiency. Based on industry requirement analysis, a comprehensive literature review, and practical experience in related fields, this research proposed a framework to utilize the digital twin for construction site management and to extend the existing LoDs of BIM for construction site management to bridge the research gaps. Extension of the LoDs' definition is the best way to be compatible with construction site management methods and meet management requirements.

This study first analyzed and outlined the components of the digital twin and indicated the operation principle and mechanism involving digital representation based on BIM, IoT, and data processing including data storage, integration, analytics, and the interaction with the physical environment. It emphasized the way the digital twin can be used in construction sites.

Second, it expanded the existing LoDs in construction sites for achieving BIM use expectation, dividing construction site components at the beginning for better illustrating the level division for graphics and information details. The LoD-Graphics and LoD-Information are divided based on the level's principle from approximate, basic, detailed, completed, and verified. BIM is able to support a number of detailed actions during construction site processing. This research considered the construction site process from inception feasibility, design authority, tender, procurement, and preconstruction (planning) to the construction stages, and then points out the minimum LoD requirement for different BIM uses in different stages.

Third, this research puts forward a solution to the problem of data management involving multiparty data management, project information management, professional technical information, data cooperation and sharing, and security and privacy issues. Furthermore, interviews and questionnaires with professionals have validated the value of this framework, complementing the problems that the framework may encounter in practice. Finally, this research indicates the future research directions about how humans can contribute to digital twins and the nontechnical factors in a project.

However, there are some limitations in this research. Because of the lack of a completed framework application case, it is difficult to clarify the improvement degree of the overall performance in the validation stage. Moreover, time and resources limited the number of interviews and questionnaires. More supplementary information can be extracted with as the number of validation participants increases.

Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

References

Abotaleb, I. S., and I. H. El-adaway. 2018. "Managing construction projects through dynamic modeling: Reviewing the existing body of knowledge and deriving future research directions." *J. Manage. Eng.* 34 (6): 04018033. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000633.

Arup. 2019. "Digital twin: Towards a meaningful framework." Accessed September 16, 2020. https://www.arup.com/perspectives/publications/research/section/digital-twin-towards-a-meaningful-framework.

Babiceanu, R. F., and R. Seker. 2016. "Big data and virtualization for manufacturing cyber-physical systems: A survey of the current status and future outlook." *Comput. Ind.* 81 (Sep): 128–137. https://doi.org/10.1016/j.compind.2016.02.004.

Barnabas, J., and P. Raj. 2020. "The human body: A digital twin of the cyber physical systems." In Vol. 117 of *Advances in computers*, 219–246. Cambridge, MA: Academic Press.

BCA (Building and Construction Authority). 2016. "BIM for DfMA (design for manufacturing and assembly) essential guide." Accessed September 16, 2020. https://www.corenet.gov.sg/media/2032999/bim_essential_guide_dfma.pdf.

Becerik-Gerber, B., F. Jazizadeh, N. Li, and G. Calis. 2012. "Application areas and data requirements for BIM-enabled facilities management." *J. Constr. Eng. Manage.* 138 (3): 431–442. https://doi.org/10.1061/(ASCE)CO.1943-7862.0000433.

Beijing Engineering Exploration and Design Association. 2013. "Building information model (BIM) design basic standards." Accessed September 16, 2020. https://wenku.baidu.com/view/fc4dc7cc69eae009581becc8.html.

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BIMForum. 2013. “Level of development specification: For building information models.” Accessed September 16, 2020. https://bimforum.org/resources/Documents/BIMForum_LOD_2013_reprint.pdf.
BIMForum. 2016. “Level of development specification.” Accessed September 16, 2020. https://bimforum.org/resources/Documents/BIMForum_LOD_2016_reprint.pdf.
BIMForum. 2019. “Level of development specification Part I & commentary for building information models and data.” Accessed September 16, 2020. https://bimforum.agc.org/wp-content/uploads/sites/27/2020/04/3.12.20-LOD-Spec-2019-Part-I-and-Guide-2019-04-29.pdf.
Bleiholder, J., and F. Naumann. 2009. “Data fusion.” *ACM Comput. Surv. (CSUR)* 41 (1): 1–41. https://doi.org/10.1145/1456650.1456651.
Bloomberg, M. R., D. Burney, and D. Resnick. 2012. “BIM guidelines.” Accessed September 16, 2020. http://www.bimfrance.net/wp-content/uploads/2013/11/NYC-BIM-Guidelines-1207.pdf.
Boje, C., A. Guerriero, S. Kubicki, and Y. Rezgui. 2020. “Towards a semantic construction digital twin: Directions for future research.” *Autom. Constr.* 114 (Jun): 103179. https://doi.org/10.1016/j.autcon.2020.103179.
Boschert, S., and R. Rosen. 2016. “Digital twin—The simulation aspect.” In *Mechatronic futures*, 59–74. New York: Springer.
Bronselaer, A., D. Van Britsom, and G. De Tré. 2014. “Propagation of data fusion.” *IEEE Trans. Knowl. Data Eng.* 27 (5): 1330–1342. https://doi.org/10.1109/TKDE.2014.2365807.
BSI (British Standard Institute). 2013. *Specification for information management for the capital/delivery phase of construction projects using building information modelling*. PAS1192-2. London: BSI.
BSI (British Standard Institute). 2014. *Specification for information management for the operational phase of assets using building information modelling*. PAS 1192-3. London: BSI.
BSI (British Standard Institute). 2015. *A specification for security-minded building information modelling, digital built environments and smart asset management*. PAS 1192-5. London: BSI.
BSI (British Standard Institute). 2018. *Specification for collaborative sharing and use of structured Health and Safety information using BIM*. PAS 1192-6. London: BSI.
Cassano, M., and M. L. Trani. 2017. “LOD standardization for construction site elements.” *Procedia Eng.* 196: 1057–1064. https://doi.org/10.1016/j.proeng.2017.08.062.
Chen, Y., and J. M. Kamara. 2011. “A framework for using mobile computing for information management on construction sites.” *Autom. Constr.* 20 (7): 776–788. https://doi.org/10.1016/j.autcon.2011.01.002.
CIC (Construction Industry Council). 2019a. “CIC building information modelling standards for mechanical electrical and plumbing.” Accessed September 16, 2020. https://bim.cic.hk/en/resources/publications_detail/66?back=%2fen%2fresources%2fpublications%3fsorting%3dseq%26keyword%3d&sorting=seq&keyword=.
CIC (Construction Industry Council). 2019b. “CIC building information modelling standards for underground utilities.” Accessed September 16, 2020. http://cloud.hkacid.com/km/assets/cic-bim-standards-for-underground-utilities.pdf.
CIC (Construction Industry Council). 2019c. “CIC production of BIM object guide general requirements.” Accessed September 16, 2020. https://www.bim.cic.hk/Upload/publication/65/download_3/271698598c1f4cc4b2ad3efd8c7bb7b4.pdf.
CIC (Construction Industry Council). 2019d. “The CIC building information modeling standards general.” Accessed September 16, 2020. https://www.bim.cic.hk/Upload/publication/68/download_3/9f00a49a3a8e42aba7ea4846932ca60f.pdf.
Cimino, C., E. Negri, and L. Fumagalli. 2019. “Review of digital twin applications in manufacturing.” *Comput. Ind.* 113 (Dec): 103130. https://doi.org/10.1016/j.compind.2019.103130.
Davis, D. 2003. “BIM (building information modeling) update.” Accessed September 16, 2020. https://www.andrew.cmu.edu/user/ramesh/teaching/course/48-749/readings/AIA%20Home.pdf.
Ding, D., Q.-L. Han, Y. Xiang, X. Ge, and X.-M. Zhang. 2018. “A survey on security control and attack detection for industrial cyber-physical systems.” *Neurocomputing* 275 (Jan): 1674–1683. https://doi.org/10.1016/j.neucom.2017.10.009.
El Amine, C. M., and O. Mohamed. 2013. “A localization and an identification system of personnel in areas at risk using a wireless sensor network.” In *Proc., 2013 Int. Conf. on Technological Advances in Electrical, Electronics and Computer Engineering (TAEECE)*, 127–131. New York: IEEE. https://doi.org/10.1109/TAEECE.2013.6557208.
Fan, C., Y. Jiang, and A. Mostafavi. 2020. “Social sensing in disaster city digital twin: Integrated textual–visual–geo framework for situational awareness during built environment disruptions.” *J. Manage. Eng.* 36 (3): 04020002. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000745.
Ford, D. N., and C. M. Wolf. 2020. “Smart cities with digital twin systems for disaster management.” *J. Manage. Eng.* 36 (4): 04020027. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000779.
Francisco, A., N. Mohammadi, and J. E. Taylor. 2020. “Smart city digital twin–enabled energy management: Toward real-time urban building energy benchmarking.” *J. Manage. Eng.* 36 (2): 04019045. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000741.
Garg, V., and N. K. Bansal. 2000. “Smart occupancy sensors to reduce energy consumption.” *Energy Build.* 32 (1): 81–87. https://doi.org/10.1016/S0378-7788(99)00040-7.
Glaessgen, E., and D. Stargel. 2012. “The digital twin paradigm for future NASA and US Air Force vehicles.” In *Proc., 53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conf. 20th AIAA/ASME/AHS Adaptive Structures Conf. 14th AIAA*, 1818. Reston, VA: American Institute of Aeronautics and Astronautics. https://doi.org/10.2514/6.2012-1818.
Gröger, G., and L. Plümer. 2012. “CityGML–Interoperable semantic 3D city models.” *ISPRS J. Photogramm. Remote Sens.* 71 (Jul): 12–33. https://doi.org/10.1016/j.isprsjprs.2012.04.004.
Guan, X., B. Yang, C. Chen, W. Dai, and Y. Wang. 2016. “A comprehensive overview of cyber-physical systems: From perspective of feedback system.” *IEEE/CAA J. Autom. Sin.* 3 (1): 1–14. https://doi.org/10.1109/JAS.2016.7373757.
Hackmann, G., W. Guo, G. Yan, Z. Sun, C. Lu, and S. Dyke. 2013. “Cyber-physical codesign of distributed structural health monitoring with wireless sensor networks.” *IEEE Trans. Parallel Distrib. Syst.* 25 (1): 63–72. https://doi.org/10.1109/TPDS.2013.30.
Ham, Y., and J. Kim. 2020. “Participatory sensing and digital twin city: Updating virtual city models for enhanced risk-informed decision-making.” *J. Manage. Eng.* 36 (3): 04020005. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000748.
Kaewunruen, S., P. Rungskunroch, and J. Welsh. 2019. “A digital-twin evaluation of net zero energy building for existing buildings.” *Sustainability* 11 (1): 159. https://doi.org/10.3390/su11010159.
Kaur, M. J., V. P. Mishra, and P. Maheshwari. 2020. “The convergence of digital twin, IoT, and machine learning: Transforming data into action.” In *Digital twin technologies and smart cities*, 3–17. New York: Springer.
Kendzierskyj, S., H. Jahankhani, A. Jamal, and J. I. Jimenez. 2019. “The transparency of big data, data harvesting and digital twins.” In *Blockchain and clinical trial*, 139–148. New York: Springer.
Kim, S., S. Chang, and D. Castro-Lacouture. 2020. “Dynamic modeling for analyzing impacts of skilled labor shortage on construction project management.” *J. Manage. Eng.* 36 (1): 04019035. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000720.
Kumar, S. S., and J. C. Cheng. 2015. “A BIM-based automated site layout planning framework for congested construction sites.” *Autom. Constr.* 59 (Nov): 24–37. https://doi.org/10.1016/j.autcon.2015.07.008.
Lee, G., J. Cho, S. Ham, T. Lee, G. Lee, S.-H. Yun, and H.-J. Yang. 2012. “A BIM-and sensor-based tower crane navigation system for blind lifts.” *Autom. Constr.* 26 (Oct): 1–10. https://doi.org/10.1016/j.autcon.2012.05.002.
Lin, J.-R., J.-P. Zhang, X.-Y. Zhang, and Z.-Z. Hu. 2019. “Automating closed-loop structural safety management for bridge construction through multisource data integration.” *Adv. Eng. Software* 128 (Feb): 152–168. https://doi.org/10.1016/j.advengsoft.2018.11.013.
Liu, Z., N. Meyendorf, and N. Mrad. 2018. “The role of data fusion in predictive maintenance using digital twin.” In *Proc., American Institute of Physics Conf.*, 020023. College Park, MD: American Institute of Physics. https://doi.org/10.1063/1.5031520.

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Lu, Q., L. Chen, S. Li, and M. Pitt. 2020a. "Semi-automatic geometric digital twinning for existing buildings based on images and CAD drawings." *Autom. Constr.* 115 (Jul): 103183. https://doi.org/10.1016/j.autcon.2020.103183.

Lu, Q., X. Xie, A. K. Parlikad, and J. M. Schooling. 2020b. "Digital twin-enabled anomaly detection for built asset monitoring in operation and maintenance." *Autom. Constr.* 118 (Oct): 103277. https://doi.org/10.1016/j.autcon.2020.103277.

Lu, Y., C. Liu, I. Kevin, K. Wang, H. Huang, and X. Xu. 2020c. "Digital Twin-driven smart manufacturing: Connotation, reference model, applications and research issues." *Rob. Comput. Integr. Manuf.* 61 (Feb): 101837. https://doi.org/10.1016/j.rcim.2019.101837.

Lydon, G. P., S. Caranovic, I. Hischier, and A. Schlueter. 2019. "Coupled simulation of thermally active building systems to support a digital twin." *Energy Build.* 202 (Nov): 109298. https://doi.org/10.1016/j.enbuild.2019.07.015.

Ma, X., F. Xiong, T. O. Olawumi, N. Dong, and A. P. Chan. 2018. "Conceptual framework and roadmap approach for integrating BIM into lifecycle project management." *J. Manage. Eng.* 34 (6): 05018011. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000647.

Ma, Z., and J. Ma. 2017. "Formulating the application functional requirements of a BIM-based collaboration platform to support IPD projects." *KSCE J. Civ. Eng.* 21 (6): 2011–2026. https://doi.org/10.1007/s12205-017-0875-4.

Miorandi, D., S. Sicari, F. De Pellegrini, and I. Chlamtac. 2012. "Internet of things: Vision, applications and research challenges." *Ad Hoc Networks* 10 (7): 1497–1516. https://doi.org/10.1016/j.adhoc.2012.02.016.

NBIMS-US (National Building Information Modeling Standard-US). 2015. *National BIM standard—United States Version 3*. Washington, DC: National Institute of Building Sciences.

Ning, X., K.-C. Lam, and M. C.-K. Lam. 2011. "A decision-making system for construction site layout planning." *Autom. Constr.* 20 (4): 459–473. https://doi.org/10.1016/j.autcon.2010.11.014.

Ozturk, G. B. 2020. "Interoperability in building information modeling for AECO/FM industry." *Autom. Constr.* 113 (May): 103122. https://doi.org/10.1016/j.autcon.2020.103122.

Panushev, I., C. Eastman, R. Sacks, M. Venugopal, and S. Aram. 2010. "Development of the national BIM standard (NBIMS) for precast/prestressed concrete." In *Proc., CIB W78 2010: 27th Int. Conf.* Ottawa: International Council for Research and Innovation in Building and Construction.

Park, K. T., D. Lee, and S. Do Noh. 2020. "Operation procedures of a work-center-level digital twin for sustainable and smart manufacturing." *Int. J. Precision Eng. Manuf.-Green Technol.* 7 (3): 791–814. https://doi.org/10.1007/s40684-020-00227-1.

Ramya, R., K. Venugopal, S. Iyengar, and L. Patnaik. 2017. "Feature extraction and duplicate detection for text mining: A survey." *Global J. Comput. Sci. Technol.* 16 (5).

Schleich, B., N. Anwer, L. Mathieu, and S. Wartzack. 2017. "Shaping the digital twin for design and production engineering." *CIRP Ann.* 66 (1): 141–144. https://doi.org/10.1016/j.cirp.2017.04.040.

Sheng, Z., S. Yang, Y. Yu, A. V. Vasilakos, J. A. McCann, and K. K. Leung. 2013. "A survey on the IETF protocol suite for the internet of things: Standards, challenges, and opportunities." *IEEE Wireless Commun.* 20 (6): 91–98. https://doi.org/10.1109/MWC.2013.6704479.

Stankovic, J. A. 2014. "Research directions for the internet of things." *IEEE Internet Things J.* 1 (1): 3–9. https://doi.org/10.1109/JIOT.2014.2312291.

Su, Y., C. Mao, R. Jiang, G. Liu, and J. Wang. 2021. "Data-driven fire safety management at building construction sites: Leveraging CNN." *J. Manage. Eng.* 37 (2): 04020108. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000877.

Tao, F., J. Cheng, Q. Qi, M. Zhang, H. Zhang, and F. Sui. 2018. "Digital twin-driven product design, manufacturing and service with big data." *Int. J. Adv. Manuf. Technol.* 94 (9–12): 3563–3576. https://doi.org/10.1007/s00170-017-0233-1.

Tolmer, C.-E., C. Castaing, Y. Diab, and D. Morand. 2017. "Adapting LOD definition to meet BIM uses requirements and data modeling for linear infrastructures projects: Using system and requirement engineering." *Visual. Eng.* 5 (1): 21. https://doi.org/10.1186/s40327-017-0059-9.

Trani, M. L., M. Cassano, D. Todaro, and B. Bossi. 2015. "BIM level of detail for construction site design." *Procedia Eng.* 123: 581–589. https://doi.org/10.1016/j.proeng.2015.10.111.

Vasenev, A., T. Hartmann, and A. G. Doree. 2014. "A distributed data collection and management framework for tracking construction operations." *Adv. Eng. Inf.* 28 (2): 127–137. https://doi.org/10.1016/j.aei.2014.01.003.

Wang, L., M. Törngren, and M. Onori. 2015. "Current status and advancement of cyber-physical systems in manufacturing." *J. Manuf. Syst.* 37 (Part 2): 517–527. https://doi.org/10.1016/j.jmsy.2015.04.008.

Wang, P., L. T. Yang, J. Li, J. Chen, and S. Hu. 2019. "Data fusion in cyber-physical-social systems: State-of-the-art and perspectives." *Inf. Fusion* 51 (Nov): 42–57. https://doi.org/10.1016/j.inffus.2018.11.002.

Whitmore, A., A. Agarwal, and L. Da Xu. 2015. "The Internet of Things—A survey of topics and trends." *Inf. Syst. Front.* 17 (2): 261–274. https://doi.org/10.1007/s10796-014-9489-2.

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